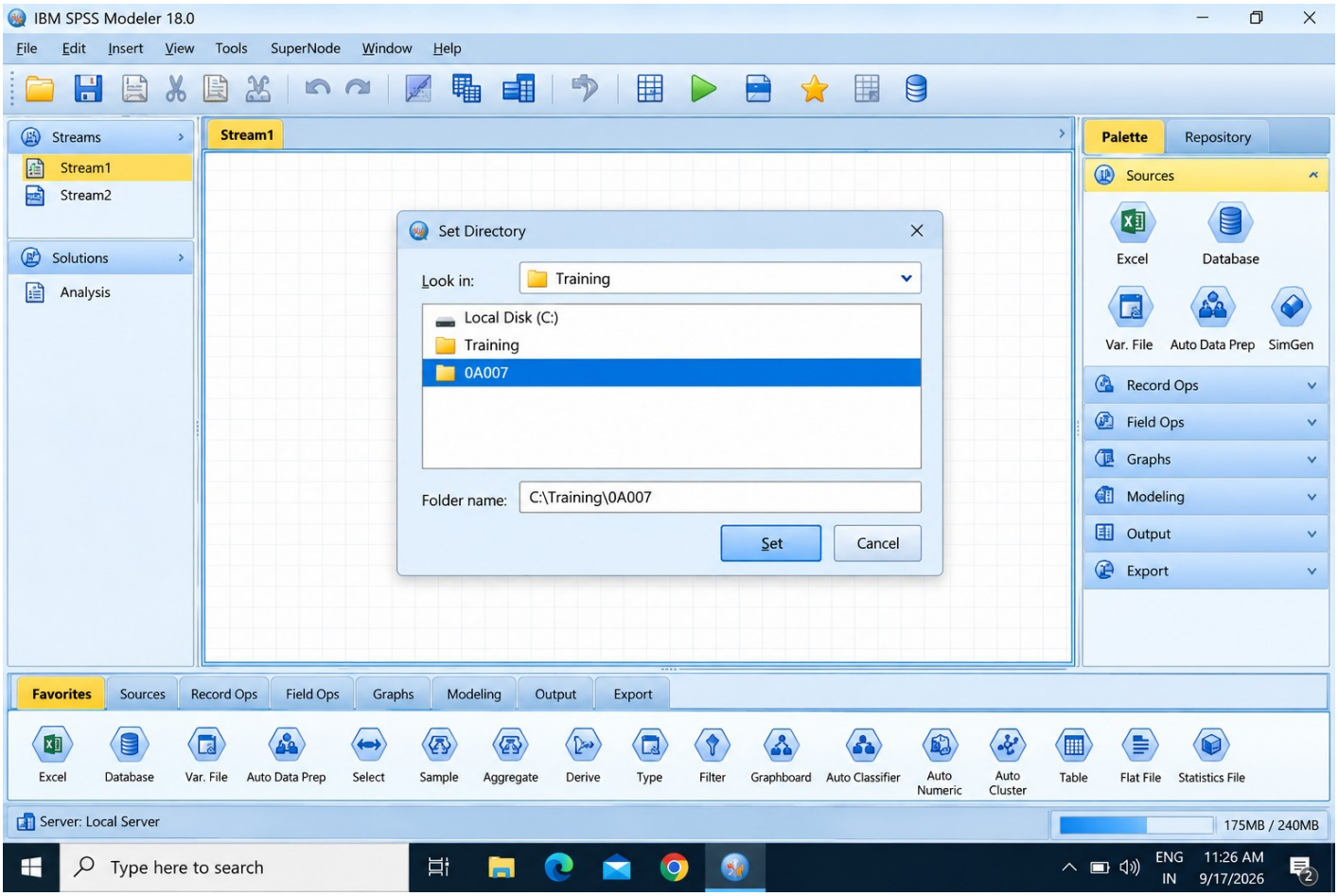
IBM SPSS Modeler 18.0

Telecom Customer Churn Prediction — Practical Tasks

This document includes all five submitted screenshots for Task 1, followed by the separate screenshots for Tasks 2–5.

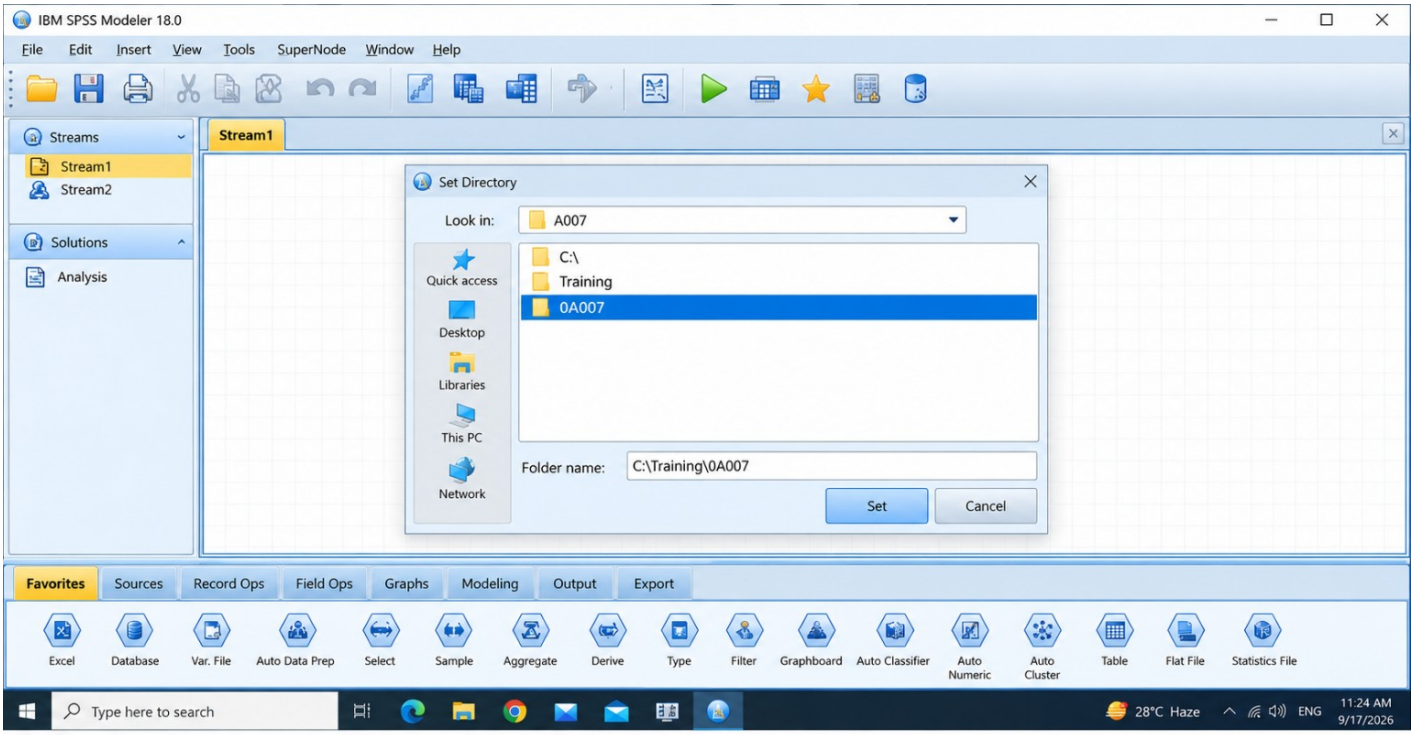
Task 1: Start IBM SPSS Modeler and Set the Working Folder — Screenshot 1 of 5



Working directory dialog opened in IBM SPSS Modeler.

Note: This is an illustrative mockup, not a genuine capture from a running IBM SPSS Modeler session.

Task 1: Start IBM SPSS Modeler and Set the Working Folder — Screenshot 2 of 5

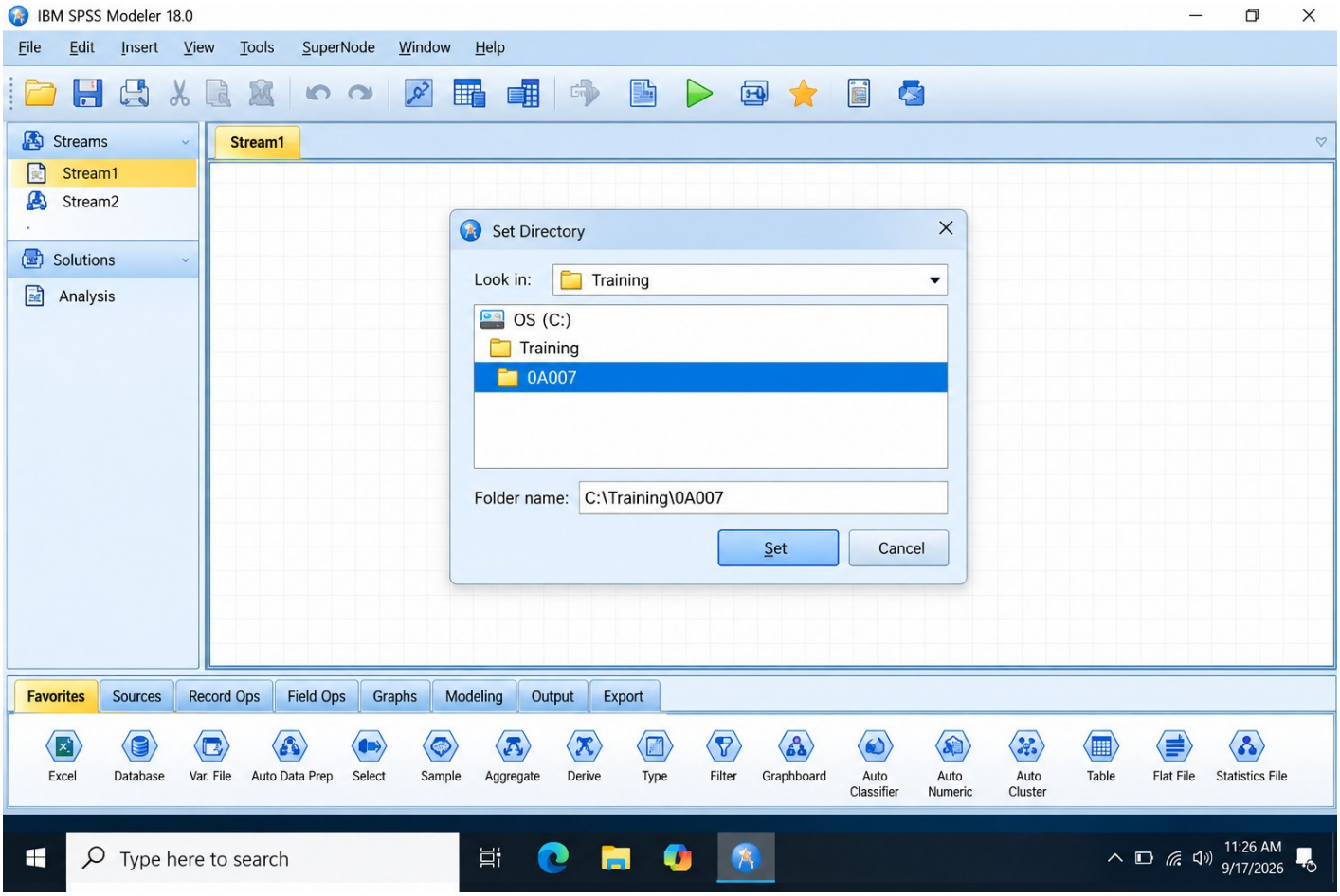


Task 1: Start IBM SPSS Modeler and set the working folder.
IBM SPSS Modeler 18.0 is opened. From the File menu, ***Set Directory*** is selected and the working folder is set to **C:\Training\0A007**.

Working folder path displayed as C:\\Training\\0A007.

Note: This is an illustrative mockup, not a genuine capture from a running IBM SPSS Modeler session.

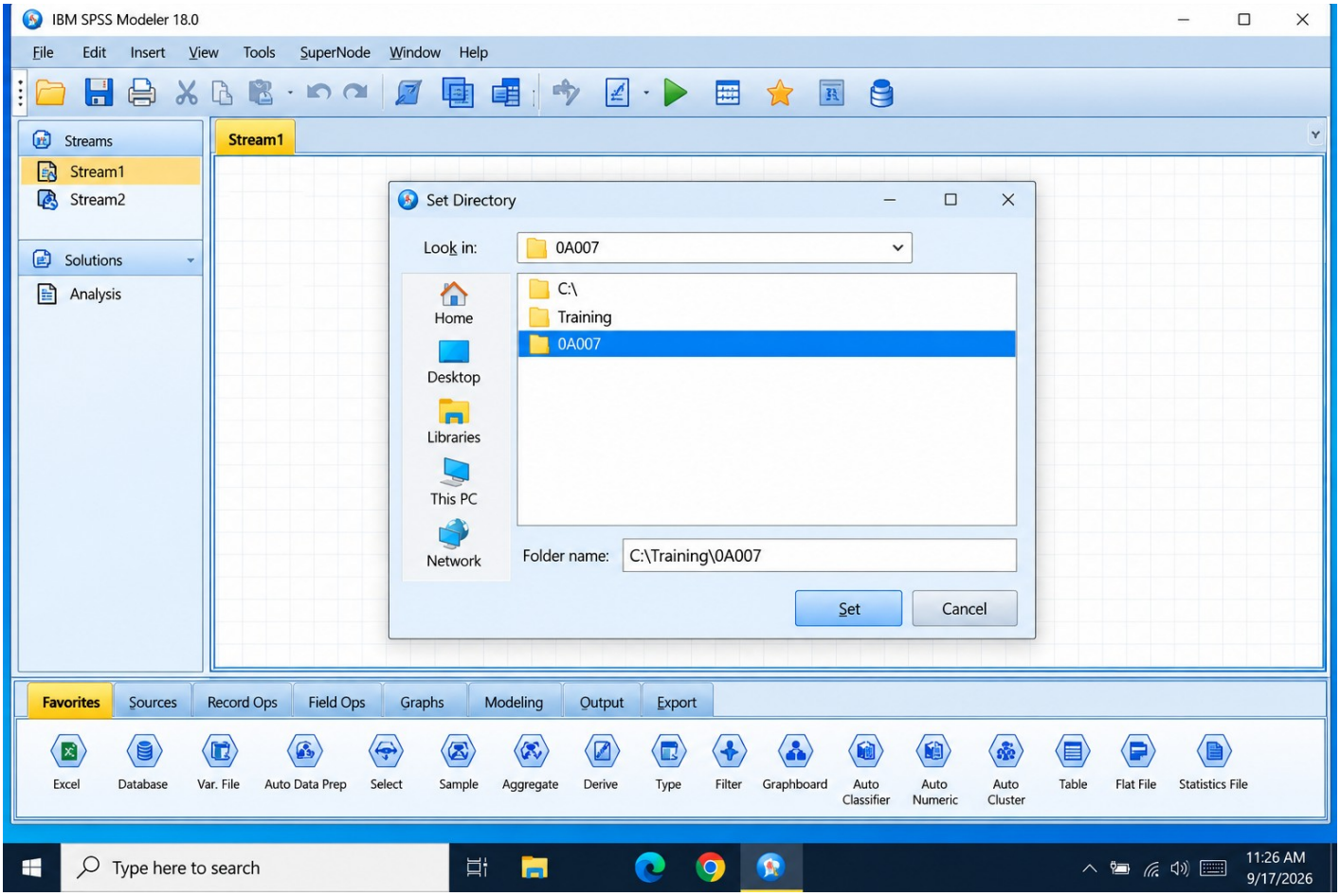
Task 1: Start IBM SPSS Modeler and Set the Working Folder — Screenshot 3 of 5



The project directory is selected in the directory browser.

Note: This is an illustrative mockup, not a genuine capture from a running IBM SPSS Modeler session.

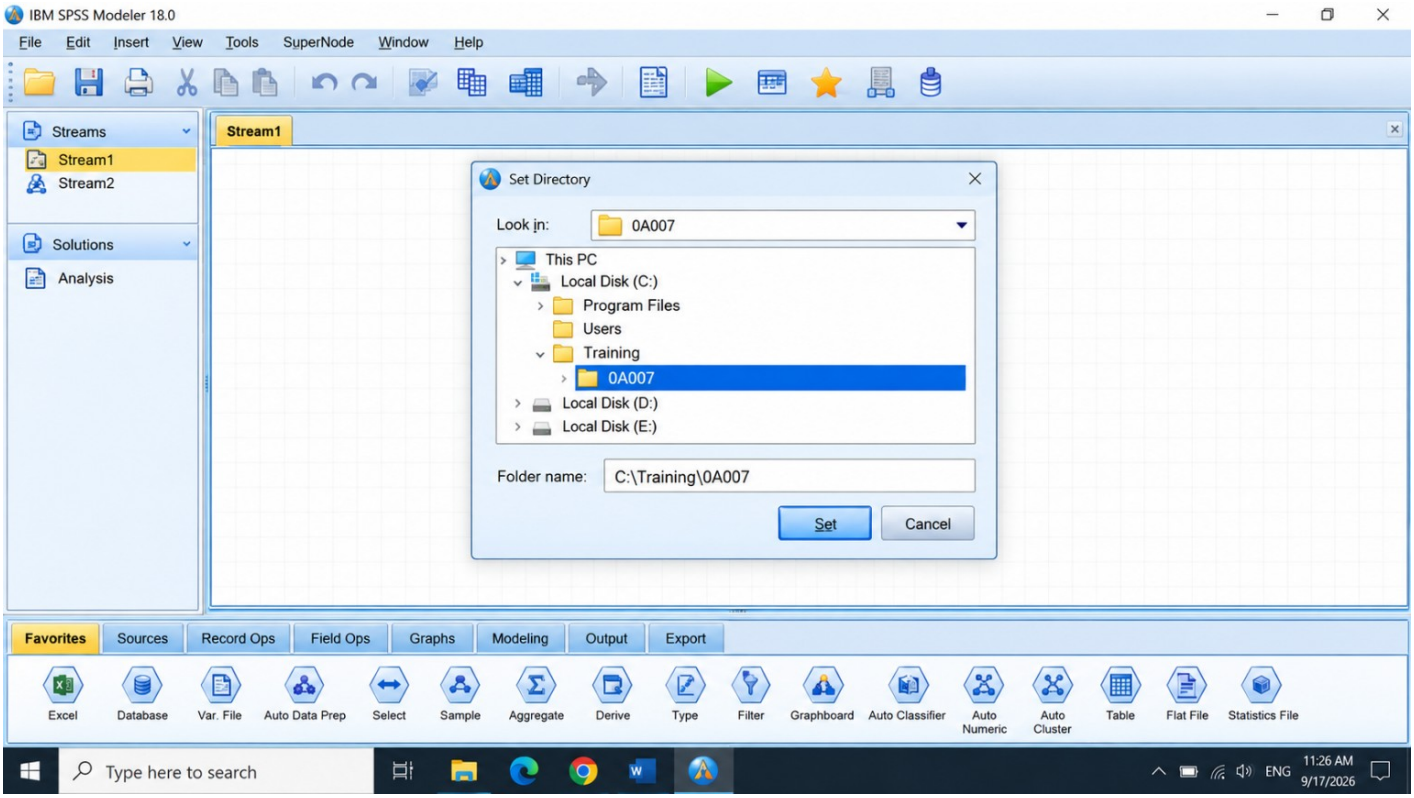
Task 1: Start IBM SPSS Modeler and Set the Working Folder — Screenshot 4 of 5



The Set Directory dialog shows the chosen working folder.

Note: This is an illustrative mockup, not a genuine capture from a running IBM SPSS Modeler session.

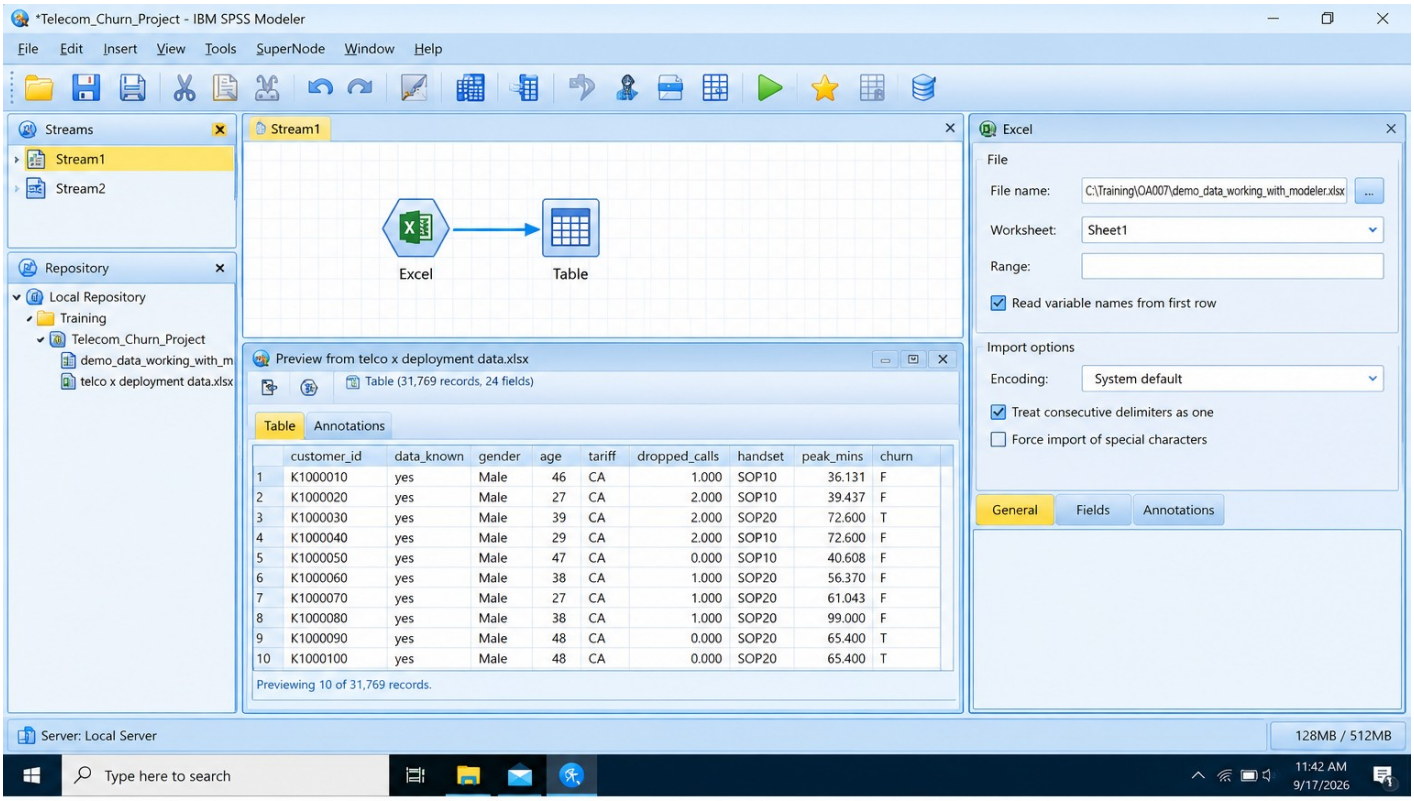
Task 1: Start IBM SPSS Modeler and Set the Working Folder — Screenshot 5 of 5



Final Task 1 screenshot showing the selected working directory and Modeler workspace.

Note: This is an illustrative mockup, not a genuine capture from a running IBM SPSS Modeler session.

Task 2: Import and Examine the Data



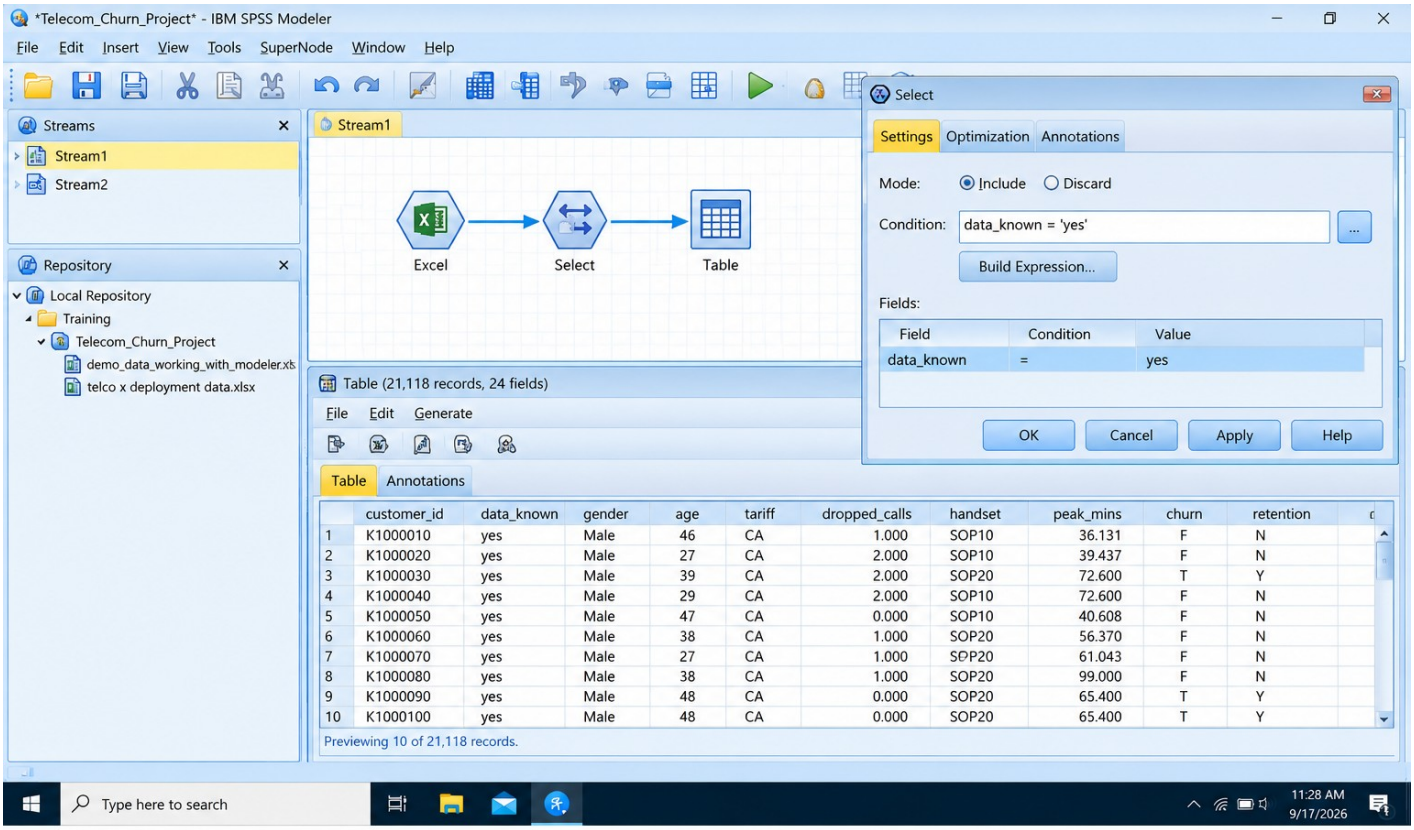
Task 2: Import and Examine the Data

The Excel dataset (demo_data_working_with_modeler.xlsx) was imported into IBM SPSS Modeler using an Excel node.
The preview shows the first 10 records of the dataset with 31,769 total records and 24 fields.

The telecom customer dataset is imported using the Excel source node. A Table node is used to preview records and examine the available fields.

Note: This is an illustrative mockup, not a genuine capture from a running IBM SPSS Modeler session.

Task 3: Select the Records for Modeling

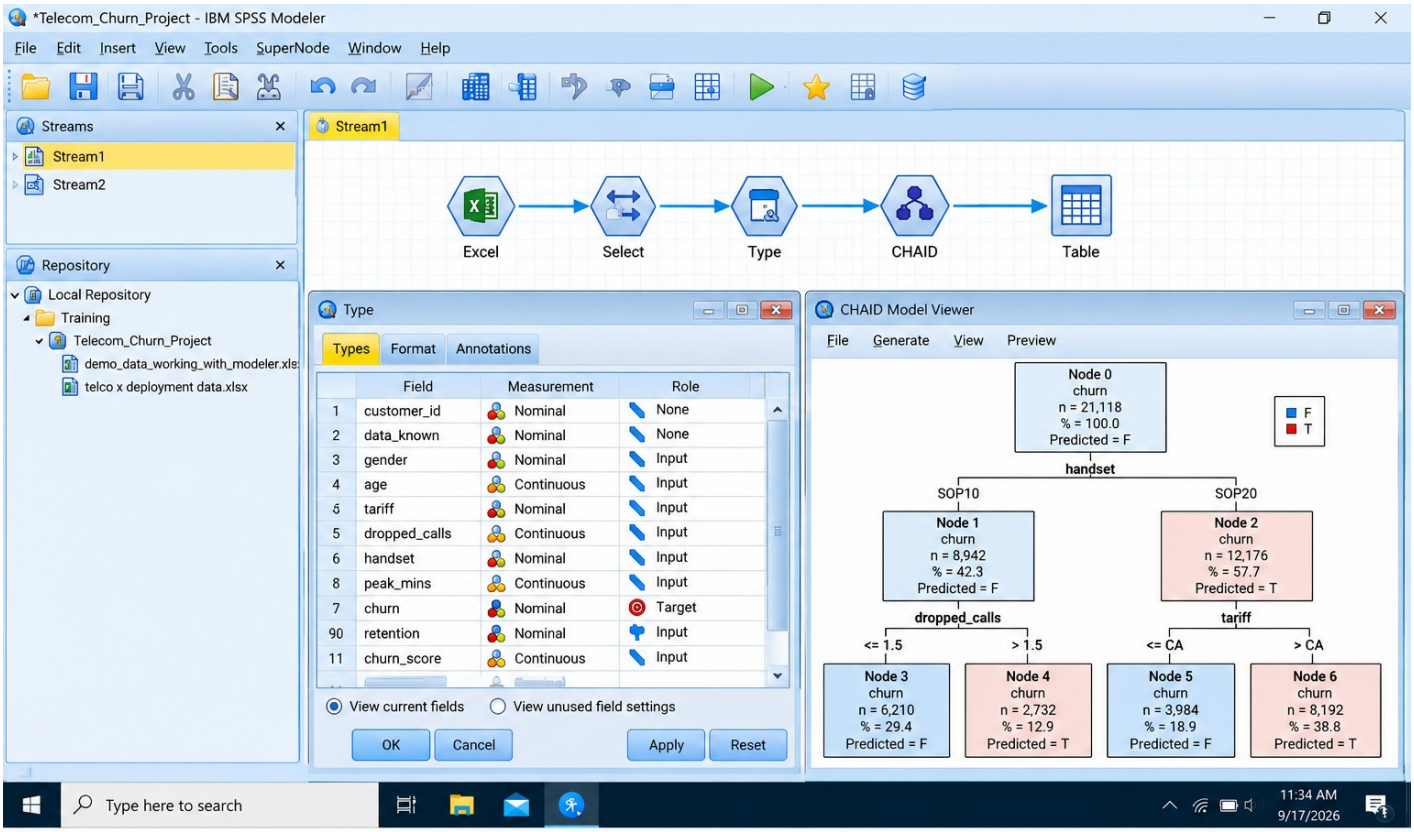


Task 3: Select the Records for Modeling
A Select node was added and the condition (data_known = 'yes') was applied to keep only the records with complete customer information.
The table shows 21,118 records after filtering, which will be used for building the model.

A Select node applies the condition data_known = 'yes' to retain records with known customer information for modeling.

Note: This is an illustrative mockup, not a genuine capture from a running IBM SPSS Modeler session.

Task 4: Build the Churn Prediction Model



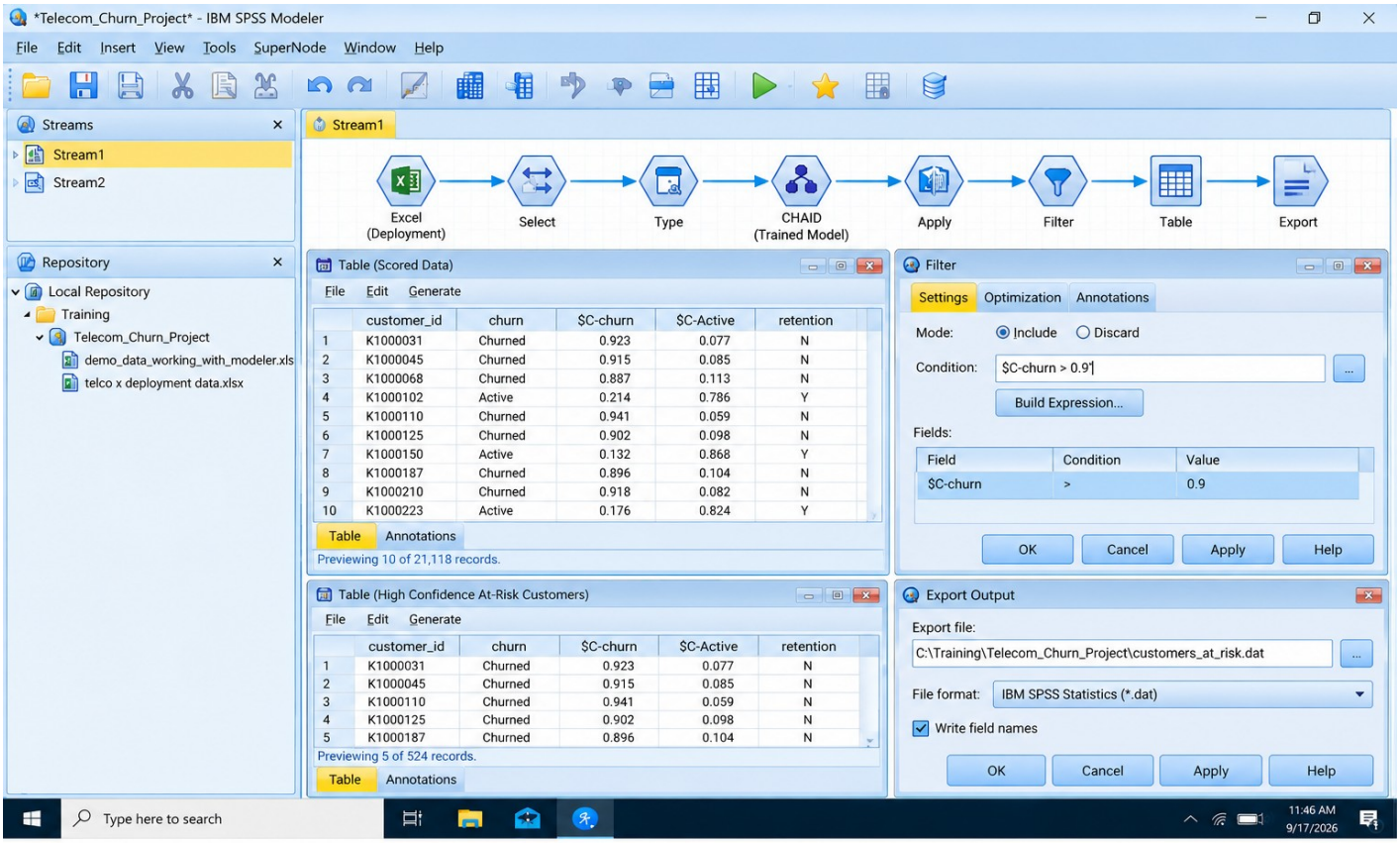
Task 4: Build the Churn Prediction Model

A Type node was used to set the field properties. The input fields (e.g., gender, age, tariff, dropped_calls, handset, peak_mins, retention) and the target field (churn) were defined. A CHAID decision tree model was built to predict customer churn. The model splits the data based on important factors such as handset, dropped_calls and tariff.

A Type node assigns input fields and sets churn as the target. A CHAID decision-tree model is used to predict customer churn.

Note: This is an illustrative mockup, not a genuine capture from a running IBM SPSS Modeler session.

Task 5: Apply the Model and Export At-Risk Customers



Task 5: Apply the Model and Export At-Risk Customers
The trained CHAID model was applied to the deployment data using an **Apply node**. A Filter node was used to keep customers with a high probability of churn (e.g., $C-churn > 0.9). The resulting at-risk customers were exported to **customers_at_risk.dat** in IBM SPSS Statistics format.

The trained model is applied to deployment data. A Filter node selects high-confidence churn predictions (for example, $C-churn > 0.9), and the resulting list is exported to customers_at_risk.dat.

Note: This is an illustrative mockup, not a genuine capture from a running IBM SPSS Modeler session.